Youwei Xiao

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Education

Bachelor of Science degree in Computer Science September 2018 - July 2022
School of Electronics Engineering and Computer Science, Peking University Beijing, China

Ph.D. Candidate in EDA September 2022 - now
School of Integrated Circuits, Peking University Beijing, China

- Research: agents for LLM infrastructure on emerging AI chips, grounded in compiler-driven hardware-software co-design.

Work Experience

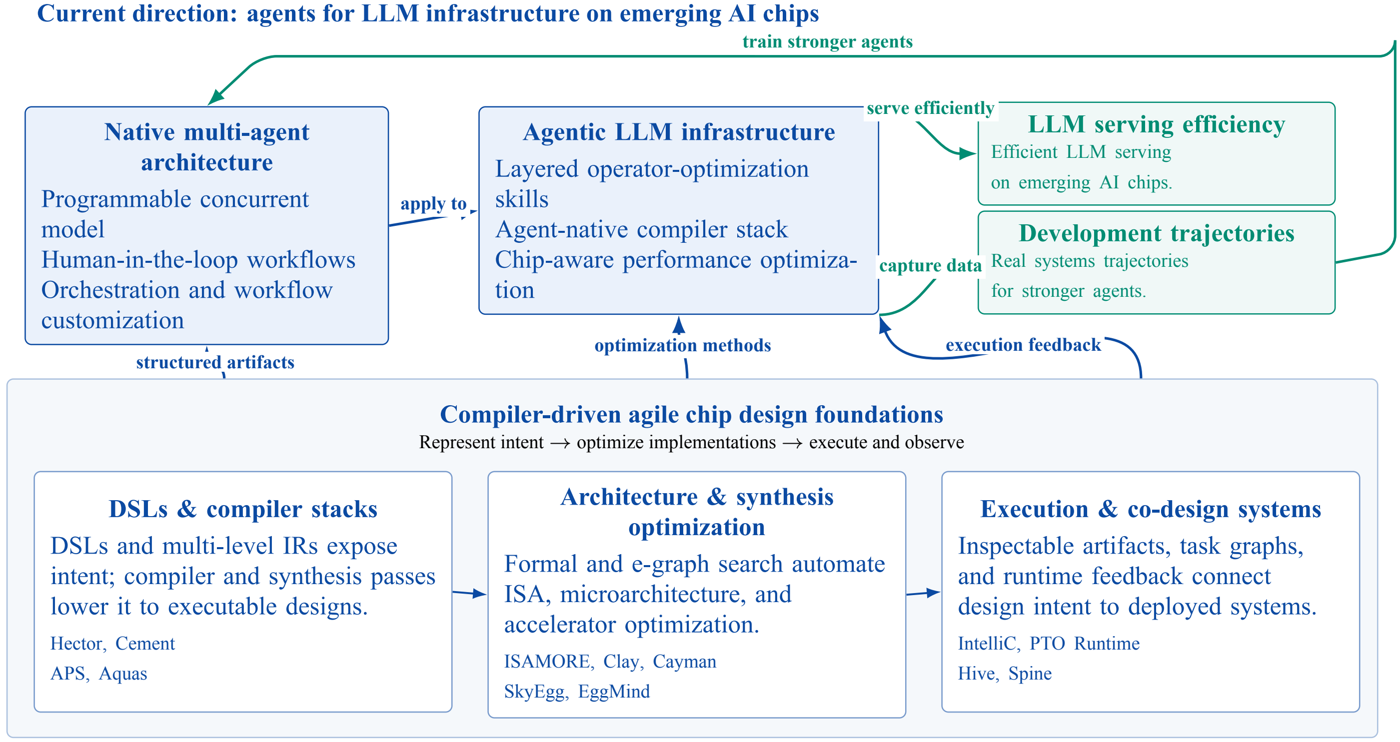
Large Language Model Algorithm Intern 2026 – Present
ByteDance Seed Beijing, China

- Contributed to native multi-agent coding systems, focusing on orchestration strategies and workflow customization with programmable concurrency and human-in-the-loop workflows.
- Applied these approaches to LLM infrastructure by evolving an abstraction-layered operator-optimization skill system and an agent-native compiler stack for emerging AI chips.
- Curated agent development data and trajectories from real systems tasks, turning infrastructure work into training signals for stronger coding and systems agents.

Research

My current research centers on agents for LLM infrastructure on emerging AI chips. I contribute to native multi-agent coding systems by focusing on orchestration strategies and workflow customization through programmable concurrency and human-in-the-loop workflows. I apply these approaches by evolving an abstraction-layered operator-optimization skill system and an agent-native compiler stack, improving LLM serving efficiency while turning real development trajectories into training data for stronger coding and systems agents.

Current direction: agents for LLM infrastructure on emerging AI chips



Compiler and runtime substrates. Earlier systems work provides the surfaces that agents can operate on. *IntelliC* (GitHub) studies inspectable compiler representations for human-agent collaboration. *PTO Runtime* (GitHub) executes compiled task graphs for serving on Ascend chips and LingQu SuperPods. *Hive* (preprint) exposes multi-agent LLM workloads to inference infrastructure, while *Spine* (GitHub) studies bounded agentic co-design across design intent, architecture, compiler, hardware, and execution evidence.

Compiler-driven agile chip design. My earlier research treats compilers as the organizing layer for architecture customization and hardware-software co-design. *Hector* (DOI, GitHub) provides multi-level IRs for hardware synthesis; *Cement* (DOI) couples the cycle-deterministic CmtHDL DSL with the CmtC timing-analysis and control-synthesis compiler; *APS* (DOI, GitHub) builds an open-source MLIR stack from architecture DSLs to hardware synthesis and software compilation, and *Aquas* (preprint) extends it with domain-specific memory/synthesis directives and an e-graph retargetable compiler. *Clay* (DOI), *Cayman* (DOI), and *SkyEgg* (preprint) automate microarchitecture and accelerator optimization, while *ISAMORE* (DOI) and *EggMind* (preprint) structure compiler search with formal and LLM-guided methods.

The resulting line of work uses earlier compiler and hardware systems as foundations for the next step: agents that help new chips serve LLM workloads, while the generated development data and trajectories feed back into stronger coding and systems models.

Publications

- **Youwei Xiao**, Chenyun Yin, Yitian Sun, and Yun Liang. 2026. Finding Reusable Instructions via E-Graph Anti-Unification. In Proceedings of the 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS ’26). doi:10.1145/3779212.3790162. **[Best Paper Award, 5/1048]**

- **Youwei Xiao**, Yuyang Zou, Yansong Xu, Yuhao Luo, Yitian Sun, Chenyun Yin, Ruifan Xu, Renze Chen, and Yun Liang. 2025. APS: Open-Source Hardware-Software Co-Design Framework for Agile Processor Specialization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD ’25). doi:10.1109/ICCAD66269.2025.11240817. [Invited Paper]

- Weijie Peng*, **Youwei Xiao***, Yuyang Zou, Zizhang Luo, and Yun Liang. 2025. Clay: High-level ASIP Framework for Flexible Microarchitecture-Aware Instruction Customization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD ’25). doi:10.1109/ICCAD66269.2025.11240669. (*Equal contribution)

- **Youwei Xiao**, Fan Cui, Zizhang Luo, Weijie Peng, and Yun Liang. 2025. Cayman: Custom Accelerator Generation with Control Flow and Data Access Optimization. In Proceedings of the 62nd ACM/IEEE Design Automation Conference (DAC ’25). doi:10.1109/DAC63849.2025.11132875.

- **Youwei Xiao**, Zizhang Luo, and Yun Liang. 2025. cmt2: Rule-Based Hardware Description in Rust with Temporal Semantics. In 5th Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE ’25). paper.

- Fan Cui, **Youwei Xiao**, Kexing Zhou, and Yun Liang. 2025. An Empirical Comparison of LLM-based Hardware Design and High-level Synthesis. In Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA ’25). doi:10.1145/3706628.3708861.

- Fan Cui, Chenyang Yin, Kexing Zhou, **Youwei Xiao**, Guangyu Sun, Qiang Xu, Qipeng Guo, Yun Liang, Xingcheng Zhang, Demin Song, and Dahua Lin. 2024. OriGen: Enhancing RTL Code Generation with Code-to-Code Augmentation and Self-Reflection. In Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD ’24). doi:10.1145/3676536.3676830.

- **Youwei Xiao**, Zizhang Luo, Kexing Zhou, and Yun Liang. 2024. Cement: Streamlining FPGA Hardware Design with Cycle-Deterministic eHDL and Synthesis. In Proceedings of the 2024 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA ’24). doi:10.1145/3626202.3637561.

- Ruifan Xu, **Youwei Xiao**, Jin Luo, and Yun Liang. 2022. HECTOR: A Multi-Level Intermediate Representation for Hardware Synthesis Methodologies. In Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design (ICCAD ’22). doi:10.1145/3508352.3549370.

Preprints

- Chenyun Yin*, **Youwei Xiao***, Yuze Luo, Yuyang Zou, and Yun Liang. LLM-Guided Strategy Synthesis for Scalable Equality Saturation. arXiv:2604.17364. (*Equal contribution)
- Zizhang Luo, Yuhao Luo, **Youwei Xiao**, Yansong Xu, Runlin Guo, and Yun Liang. Hive: A Multi-Agent Infrastructure for Algorithm- and Task-Level Scaling. arXiv:2604.17353.
- Yuyang Zou, **Youwei Xiao**, Yansong Xu, Chenyun Yin, Yuhao Luo, Yitian Sun, Ruifan Xu, Renze Chen, and Yun Liang. Aquas: Enhancing Domain Specialization through Holistic Hardware-Software Co-Optimization based on MLIR. arXiv:2511.22267.
- **Youwei Xiao**, Yuyang Zou, and Yun Liang. SkyEgg: Joint Implementation Selection and Scheduling for Hardware Synthesis using E-graphs. arXiv:2511.15323.
- **Youwei Xiao**, Zizhang Luo, Weijie Peng, Yuyang Zou, and Yun Liang. Cement2: Temporal Hardware Transactions for High-Level and Efficient FPGA Programming. arXiv:2511.15073.

Honors and Awards

学术之芯 — 2026
Awarded to 8 students for academic excellence — School of IC, Peking University

博士研究生校长奖学金 — 2026-2027
Awarded to 10 students in School of IC, Peking University — Peking University

ASPLOS 2026 Best Paper Award — March 2026
Finding Reusable Instructions via E-Graph Anti-Unification — 5/1048 submissions

Second Place in EDAthon 2020 — August 2020
Programming Competition on Electronic Design Automation — IEEE CEDA Hong Kong Chapter

Shenzhen Stock Exchange Scholarship — 2019-2020
Awarded to 17 students from EECS, Peking University

Merit Student — 2018-2019 & 2019-2020
Awarded to top 10% students for comprehensive excellence

Yang Fuqing & Wang Yangyuan Academician Scholarship — 2018-2019
Awarded to only 5 students at EECS, Peking University, for academic excellence

Second Prize in NOI 2017 — July 2017
National Olympiad in Informatics — China Computer Federation

Second Prize in APIO 2017 — May 2017
Asia Pacific Informatics Olympiad (China District) — China Computer Federation

Tutorials

APS: An MLIR-Based Hardware-Software Co-design Framework — January 2026
ASP-DAC 2026 — Hong Kong, China

Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design — May 2025
ISEDA 2025 — Hong Kong, China

Agile Hardware Specialization (AHS) — March 2025
ASPLOS 2025 — Rotterdam, Netherlands

Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design — March 2025
DATE 2025 — Lyon, France

AHS: An EDA toolbox for Agile Chip Front-end Design — January 2025
ASP-DAC 2025 — Tokyo, Japan